

wires in the breadboard, poke the probes from the VCC pin on a micro-controller to the pin that's at the top of the column from the IC's VCC pin. A tone will be emitted which indicates that power is free to flow from the VCC pin to the breadboard. If the tone isn't emitted, remove the IC and try inserting it again properly.

Additionally, you can also check if two SMD pins are connected on an assembled PCB if they aren't in your field of view. Also, when a system is not working, continuity is a great way to troubleshoot it.

Picking a good multimeter

Let's look at some of the features we think you should look out for when buying a multimeter.

Autoranging

There are plenty of multimeters that are autoranging - they automatically change internal range to accurately measure the voltage, resistance, or current. These are usually higher quality multimeters and have more features. Additionally, you can also switch it to manual mode for times when the current or voltage is volatile and auto-range can't keep up with it.

Back-lit LCD

A back-lit LCD is useful and allows you to measure your circuit even in the dark, although not many situations like this might arise, it is still a useful feature to have.

Good click

It is preferred to have a good click or 'clicky feeling' on the range selector of your multimeter. A soft knob usually implies a lower quality meter.

Good probes

Make sure you buy probes that are decently durable. Over time, you will see that the leads will break down at the flex point and sometimes, wires come out of the probes completely. More robust probes should last longer, however, if they do break they are cheap to replace.

Auto-off feature

This feature is fairly uncommon on cheaper multimeters. However, it is one that can benefit novices and advanced users alike, since turning off the